BIG MODULAR / WIRED POWER FLOW

AUTONOMOUS 9 V INPUT / V1 — UNROUTED / PLACEMENT NEXT

USB PD input

USB-C / FIXED9V

VIN9

GND

Autonomous hardware window and retry

U2 WINDOW / SW1 RETRY

VIN9

VPROT

GND

Independent buck converters

U3 MID / U5 +5V

VPROT

MID

P5

GND

Direct bipolar rails

U4 / +/−12V

MID

GND

P12

N12

Board output and power LED

J7 / D10 P5 LED

P5

P12

N12

GND

9 V voltage acceptance is NOT proof of a PD protocol contract.
No MCU, AUX regulator, sensing/debug or artificial RUN inhibit.
SW1 retries eFuse; UV/OV window remains independent.

J7 is the physical output boundary: one manual harness.
Top USB/retry/3mm LED are placed together in the native boards.
All64 footprints synchronized:4 fixed,60 off-board staged; no copper.

V1 — UNROUTED / PLACEMENT NEXT

Big Modular

Sheet: /

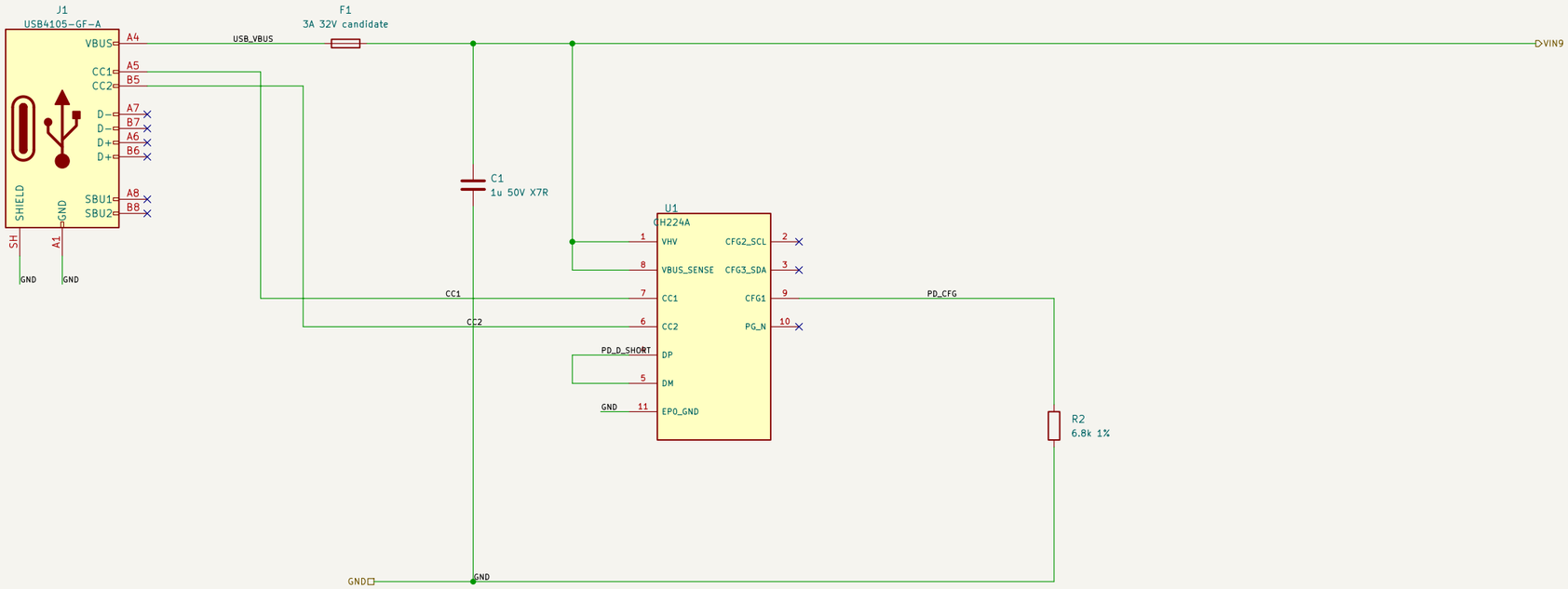
File: big_modular.kicad_sch

Title: Big Modular — V1 electrical hierarchy

Size: User	Date:	**Rev: V1 routing preparation**
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01 / USB PD INPUT

USB_VBUS -> F1 -> VIN9 (fused, may be 5 V during negotiation).
CH224A fixed 9 V via 6.8k CFG1; CFG2/3 and PG unused. DP/DM shorted locally.
VHV1 and VBUS8 directly VIN9 per WCH table4-1 / section6.1; EP0 -> footprint11.



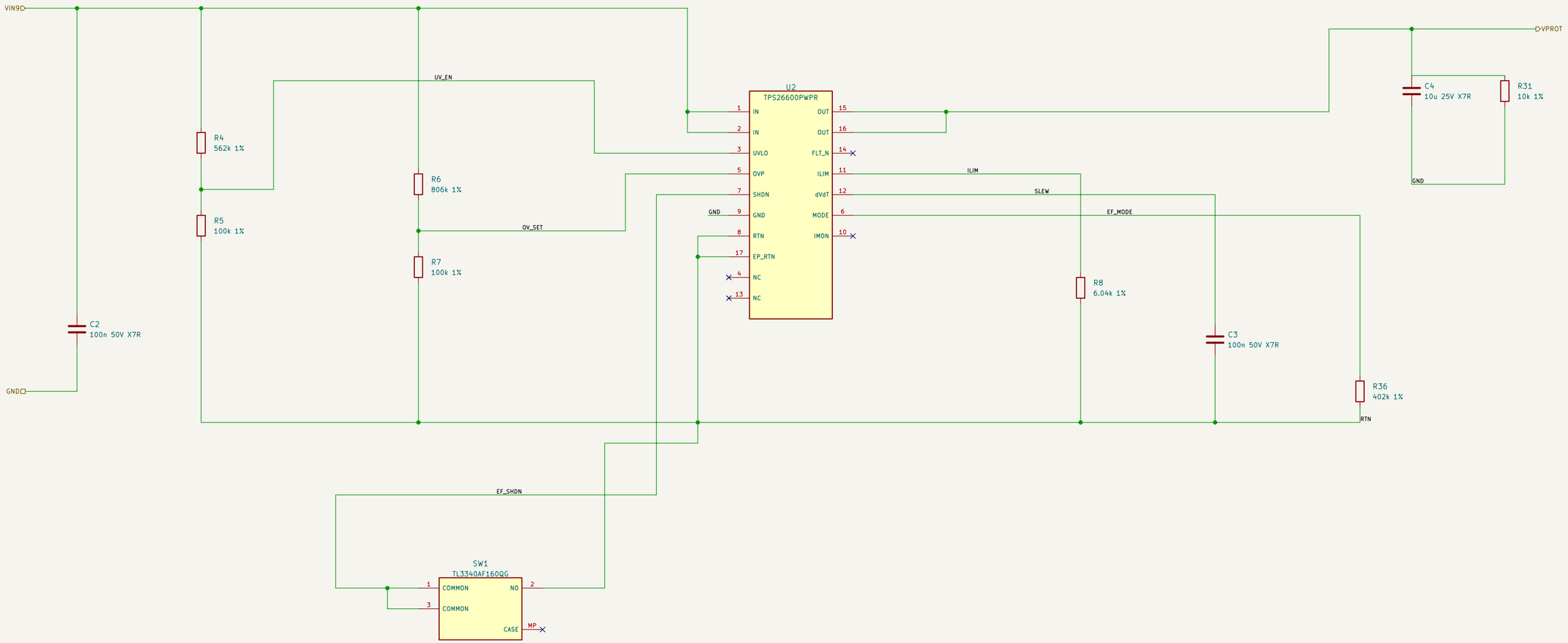
6.8k single-resistor configuration requests 9 V.
CFG2/3 internal pullups; intentionally unused (no I2C header).
PG intentionally NC: voltage window, not protocol status, permits power.



V1 — UNROUTED / PLACEMENT NEXT
Big Modular
Sheet: /USB PD input/
File: 01_input_protection.kicad_sch
Title: Big Modular — USB PD input
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02 / AUTONOMOUS HARDWARE WINDOW AND RETRY

UV rising 7.878 V / OV rising 10.781 V nominal: hysteresis applies. RTN is NOT GND.
SHDN self-biases internally; SW1 shorts SHDN to RTN while pressed, release resets/retries.
R8 6.04k: 1.987 A nominal. MODE 402k: current limit with thermal latch-off. No PG/MCU permission.



SW1 PRESS: SHDN -> RTN, eFuse OFF; RELEASE: internal 2.0..3.4 V bias retries.
No AUX/PG/JP1 or external permission. UV/OV/current/thermal protection always applies.
Bucks use their own internal EN pullups, not this weak SHDN bias.

V1 — UNROUTED / PLACEMENT NEXT
Big Modular
Sheet: /Autonomous hardware window and retry/
File: 02_hardgate.kicad_sch
Title: Big Modular — Autonomous hardware window and retry
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03 / INDEPENDENT BUCK CONVERTERS

U3 MID 5.216 V: upper DC corner 5.357 V BEFORE transients. U4 ceiling 5.5 V.
U5 direct 5.011 V from VPROT, never from positive boost. 100n BOOT-to-SW each.
EN pins intentionally floating: TI internal pullups enable from protected VPROT; no SHDN loading.

U3: MID / 5.216 V

VPROT

C35
100n 50V X7R

U3
TPS54302DDCR
VIN
EN
GND
BOOT
SW
FB

MID_BOOT
MID_SW
MID_FB

L1
6.8uH 20% shielded

MID

R9
100k 0.1%

R10
12.9k 0.1%

C5
10u 25V X7R

C7
100n 50V X7R

C6
22u 25V X7R

C37
22u 25V X7R

GND

U5: P5 / 5.011 V

C36
100n 50V X7R

U5
TPS54302DDCR
VIN
EN
GND
BOOT
SW
FB

FIVE_BOOT
FIVE_SW
FIVE_FB

L4
6.8uH 20% shielded

P5

R17
100k 1%

R18
13.5k 0.1%

C18
10u 25V X7R

C20
100n 50V X7R

C19
22u 25V X7R

C38
22u 25V X7R

GND

BOOT caps are switch-node referenced, not grounded.
Kelvin FB ground to IC GND; keep high dI/dt loops off analog return.
6.8uH Sunlord SWPA80655; validate thermal/loop behavior in V1.

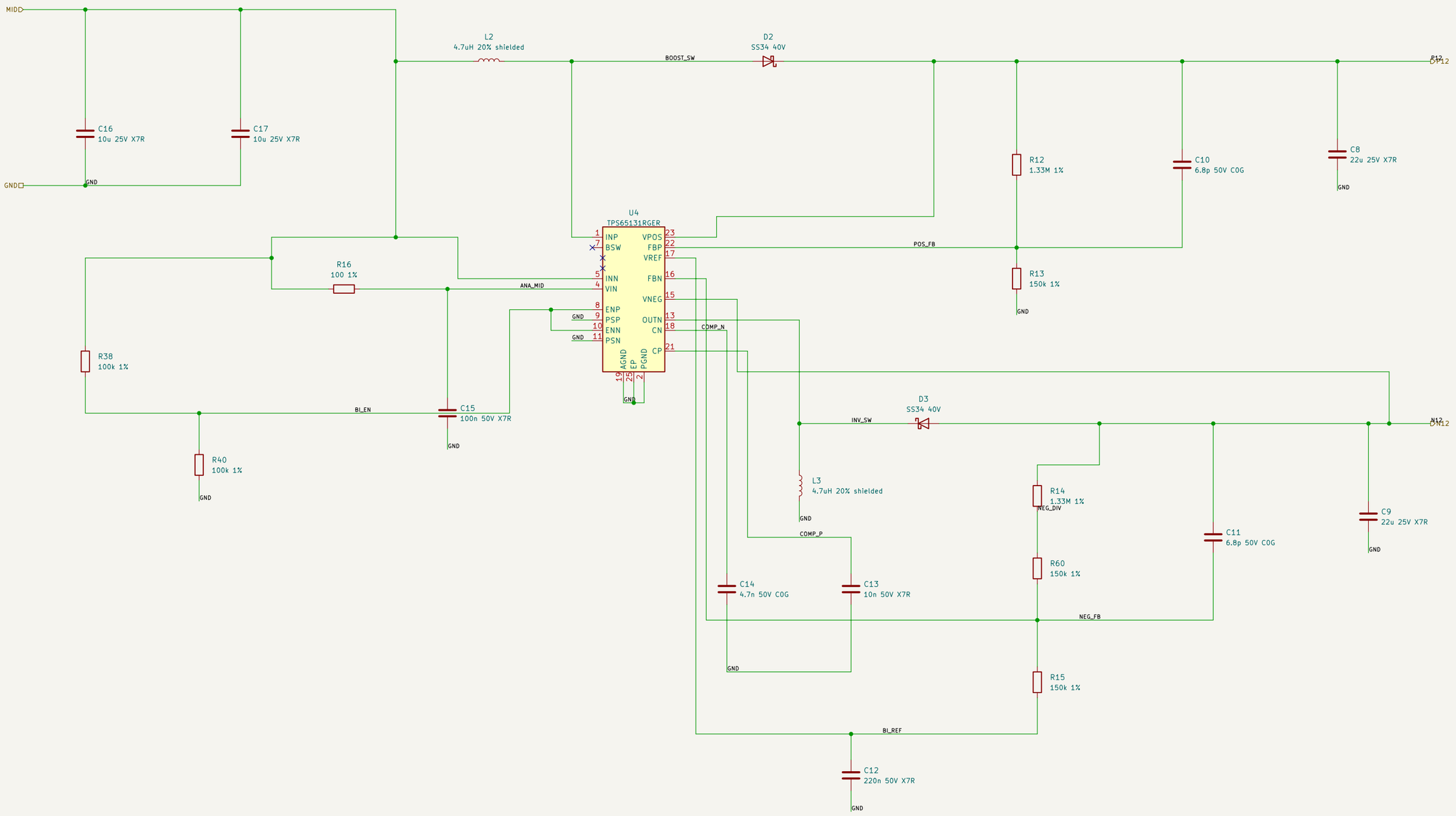
#FLG04
PWR_FLAG
MID

#FLG08
PWR_FLAG
P5

V1 – UNROUTED / PLACEMENT NEXT
Big Modular
Sheet: /Independent buck converters/
File: 03_bucks.kicad_sch
Title: Big Modular – Independent buck converters
Size: User | Date:
Rev: V1 routing preparation
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04 / DIRECT BIPOLAR RAILS

Nominal +/−11.968 V. FBN returns to VREF, NOT GND; PSP/PSN grounded.
Local output feedback: no rail LDOs or extra output LC. L2/L3 4.7uH per TI typical application.
Output current/ripple/thermal/transient limits need V1 validation; no simultaneous +/−800 mA rating.



Separate INP/INN local ceramic loops; shielded L2/L3.
No arbitrary output LC. Compensation and Cff require loop validation.

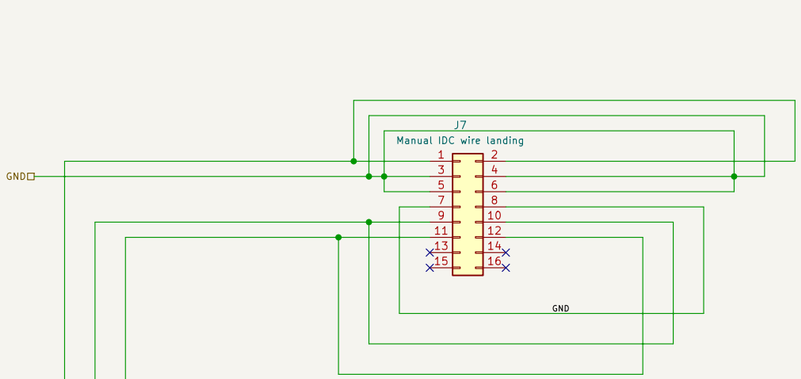
#FLG05 PWR_FLAG — ANA_MID
#FLG06 PWR_FLAG — P12
#FLG07 PWR_FLAG — N12

V1 — UNROUTED / PLACEMENT NEXT
Big Modular
Sheet: /Direct bipolar rails/
File: 04_bipolar.kicad_sch
Title: Big Modular — Direct bipolar rails
Size: User | Date: | **Rev: V1 routing preparation**
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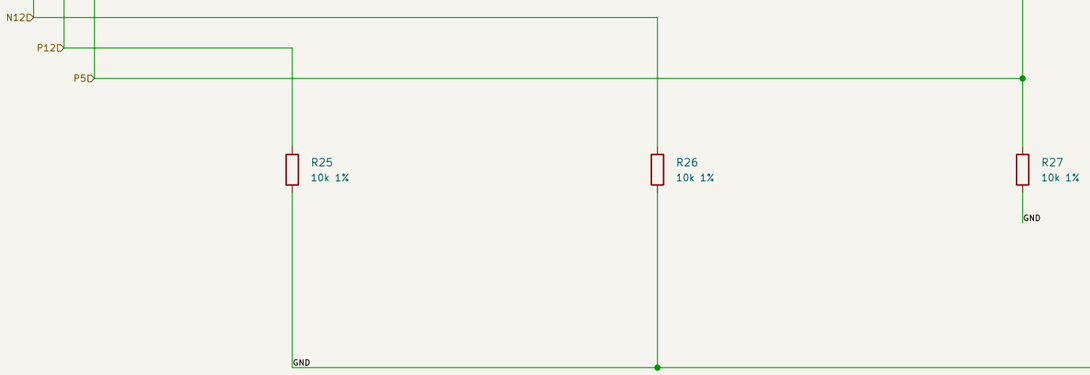
05 / BOARD OUTPUT AND POWER LED

J7 physical board wire landing: 1/2=−12, 3..8=GND, 9/10=+12, 11/12=+5; 13..16 NC.
One manual output harness, no J2 symbol or second source. Never parallel another active PSU.
Single red D10: final P5 -> 1k -> anode2; cathode1 -> GND. +5 V presence only.

J7: PHYSICAL BOARD OUTPUT LANDING







V1 — UNROUTED / PLACEMENT NEXT
Big Modular
Sheet: /Board output and power LED/
File: 07_output_rails.kicad_sch
Title: Big Modular — Board output and power LED
Size: User | Date: | **Rev: V1 routing preparation**
KiCad E.D.A. 10.0.6 | Id: 6/6